DHAKA AIR

AIR QUALITY PREDICTION SYSTEM

CSE299 Project Proposal

Track 3: ML Model & Dataset Development

Using a Mendeley air-quality dataset with support for adding new datasets later.

GROUP 8

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AQI
Predicted Level

PM2.5
Input Feature

PM10
Input Feature

Problem and Motivation

Why this project matters in the local context

Local problem

Dhaka often faces unhealthy air conditions, but raw pollution data is hard for users to understand or use for decisions.

Motivation

- People need a simple way to understand pollution trends.
- Air quality changes by time, location, and pollutant levels.
- Students/researchers need a reusable ML-based analysis system.
- New datasets should be easy to add for future improvements.

Pollution Data

PM2.5 • PM10 • CO • NO_2 • SO_2

▼

ML Analysis

Learns patterns from data

▼

Better View

Prediction + visual results

Main idea: turn complex air-quality data into readable prediction results.

Objective, Scope & Non-Triviality

What we will build and why it is technically challenging

Objective

Build a web application that predicts Dhaka air quality using ML models and shows the results with graphs, metrics, and an easy dashboard.

Scope

- Dataset preparation
- Model training & comparison
- Prediction dashboard
- New dataset upload
- What-if input testing

Non-Trivial

Real data may be missing, noisy, or imbalanced. The system must validate new datasets, retrain models, compare results, and explain model errors.

PROJECT BOUNDARY



Data and Framework

Data source, new-dataset support, and ML pipeline

Mendeley Air Quality Dataset

Start with a public Mendeley dataset for Bangladesh/Dhaka air quality. Core fields may include timestamp, city/location, PM2.5, PM10, pollutants, AQI, and coordinates.

Reusable Dataset Upload

Upload CSV/Excel → validate required columns → clean data → save as a new dataset → retrain/test models.

CSV Upload | Validation | Preprocess | ML Models | Dashboard

Columns

Required: date/time, location, PM2.5/PM10, AQI or target label.
Optional: temperature, humidity, wind speed, other pollutants.

Technology

Python • Pandas • NumPy • Scikit-learn
Flask/Django • HTML/CSS/JS • GitHub

Web Application Features

Main functions users will see in the final system

1

Dashboard

AQI, PM2.5, PM10, pollutant summary and graphs

2

Historical Viewer

Filter dataset by city, date and time period

3

Dataset Upload

Add new CSV/Excel dataset with validation

4

ML Prediction

Predict AQI value or air-quality category

5

Model Comparison

Compare baseline and multiple ML models

6

What-If Testing

Change input values and see prediction changes

7

Error Analysis

Show mistakes, metrics and confusion/errors

8

Results Export

Save graphs/results for report or presentation

User flow: Upload/select dataset → choose model → predict AQI → compare results → view dashboard.

Implementation Stages

Step-by-step development plan

1 Dataset setup

Download Mendeley data and define required columns

2 Preprocessing

Clean missing values, duplicates, formats and labels

3 Model training

Train baseline and ML prediction models

4 Evaluation

Compare models and analyze errors

5 Web app

Build dashboard, upload feature and graphs

6 Testing

Test with selected and newly uploaded datasets

Milestone focus: keep a working version early, then improve models and visuals.

Evaluation Plan and Deliverables

How we will prove the system works

Model Metrics

MAE / RMSE for AQI value; Accuracy / F1 for category

Model Comparison

Baseline vs ML models on the same test data

Robustness

Test with new uploaded dataset or changed data subset

Usability Check

Can user upload data, run prediction, and view graphs easily?

DELIVERABLES

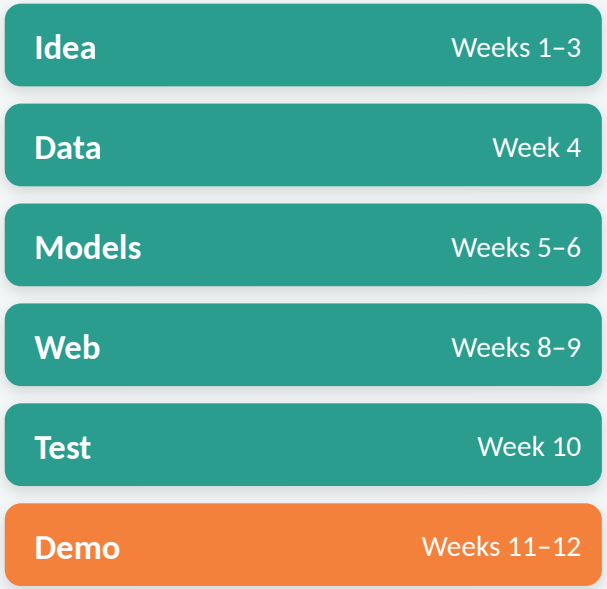
- Prepared Mendeley-based dataset
- Dataset upload and validation module
- Trained ML models with comparison report
- Prediction dashboard with graphs
- Error analysis and limitations
- Final report, slides and live demo

Success target: add/select data, run prediction, compare models, and understand the result visually.

Week-by-Week Schedule

6-minute proposal plan and development timeline

Week	Work Plan
1–2	Finalize idea + confirm dataset
3	Proposal presentation
4	Clean and preprocess data
5	Build baseline + first ML model
6	Model comparison + error analysis
7	Mid-project report
8	Dataset upload module
9	Dashboard + graphs
10	Integration + testing
11–12	Final demo, slides, report



Thank You

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Questions & feedback are welcome.

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